YTÜ –Faculty of Chemical and Metallurgical Department of Mathematical Engineering 2.Midterm Exam Question and Solution Sheet		**SCORE TABLE**					
		1	**2**	**3**	**4**	**SUM**	
Name-Surname							
Number							

Department						**Date**	**09.06.2021**
Lecture	**MTM3522 ALGEBRA**	**Grup Number**	**2**	Duration	90 min.	**Room**	Online
Lecturer	Assist. Prof. Dr. Serkan ONAR					**Signature**	

You must show all details of your work to get full credit. Correct answers with no detail will get zero credit.

1- (18pt) For cyclic group C_4 and dihedral group D_4;

i) Determine the elements.
ii) Find the order of these groups.
iii) Is C_4 and D_4 p-group?
iv) Find the index and write the cosets.
v) Show that $C_4 \lhd D_4$.
vi) Write the elements of quotient group D_4 / C_4.

2- (24pt) For groups $\mathbb{Z}_8$ and $\mathbb{Z}_9^*$;
i) Determine the elements of both groups.
ii) According to which operations, both groups are abelian ?
iii) Are these two groups are cyclic?
iv) If cyclic, then find the generators?
v) Write the elements of direct product $\mathbb{Z}_8 \times \mathbb{Z}_9^*$ and find the order.
vi) What is the unit element of group $\mathbb{Z}_8 \times \mathbb{Z}_9^*$?
vii) Find the order of element $(6,5) \in \mathbb{Z}_8 \times \mathbb{Z}_9^*$, $o(6,5) = ?$
viii) Find the inverse of element $(6,5) \in \mathbb{Z}_8 \times \mathbb{Z}_9^*$, $(6,5)^{-1} = ?$

3- (18pt) Find the subgroups $H_1 = \langle \overline{2} \rangle$, $H_2 = \langle \overline{3} \rangle$, $H_3 = \langle \overline{4} \rangle$ of group $G = \mathbb{Z}_{12}$. Which of these subgroups is written as internal direct sum?

4- (19pt) Find the Sylow-subgroups of group $G = \mathbb{Z}_{70}$. Write as direct sum.

5- (21pt) Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 7 & 5 & 4 & 3 & 2 & 1 \end{pmatrix}$ and $\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 5 & 2 & 7 & 6 & 4 & 3 \end{pmatrix}$ on S_7.

i) Write σ and ρ as a product of disjoint cycles.
ii) Determine product of $\sigma\rho$.
iii) Find the inverse of $\sigma\rho$, $(\sigma\rho)^{-1}$=? .
iv) Find the order of $\sigma\rho$.
v) Write $\sigma\rho$ as a product of transpositions.
vi) Is $\sigma\rho$ even or odd permutation.
vii) Find $\sigma\rho\sigma^{-1}$.

Exam Duration: 90 min, **Log in system and upload files:** 10 min ***Good Luck.***